

Histone Modification Profiles Reveal Dynamics of *Polycomb* And *Trithorax* Group Activities in *Arabidopsis* Embryos

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Abstract

Background: precise regulation of gene expression is a crucial prerequisite for the development of multicellular organisms, including both plants and animals. Histone modifications H3K27me3, H2Aub, and H3K4me3 function as hallmarks of the activities of *Polycomb* and *Trithorax* group proteins and represent a key class of epigenetic regulatory mechanisms. Despite their central role in developmental regulation, their dynamics in embryogenesis - an early and integral part of the plant life cycle - remain largely unexplored. Embryogenesis involves key developmental events, including phase transition, establishment and maintenance of cell fate, regulation of growth, and cell cycle progression. Thus, genome-wide profiles of histone modifications in embryos could provide valuable insights into how the *Polycomb* and *Trithorax* group proteins coordinate these events and advance our understanding of the dynamics of *Polycomb* and *Trithorax* group activities throughout the plant life cycle.

Results: We mapped the genome-wide distribution of H3K4me3, H3K27me3, and H2Aub in early and late stages of *Arabidopsis* embryo development. The results revealed that H3K4me3 is associated with growth and cell cycle progression at the early stage of embryonic development; H3K27me3 represses the seed maturation program in embryos at the early stage; and H2Aub is associated with the active expression of seed development genes in embryos.

Conclusion: Profiles of H3K4me3, H3K27me3, and H2Aub revealed dynamic *Polycomb* and *Trithorax* group activities in *Arabidopsis* embryos, highlighting their roles in regulating growth and cell proliferation, developmental phase transition, and the non-canonical association of H2Aub with active gene expression.

Keywords: Epigenome, Histone Modifications, Embryogenesis, *Polycomb* group proteins, *Trithorax* group proteins, CUT&Tag, *Arabidopsis*, Developmental Epigenetics

37 Introduction

38 Although cells in a multicellular organism share nearly identical genome sequences,
39 they give rise to diverse cell types, tissues, and developmental programs. Epigenetic
40 regulations of gene expression, independent of DNA sequence changes, play a central
41 role in establishing and maintaining this diversity [1]. A key class of epigenetic
42 regulation is mediated by histone modifications, which are chemical modifications of
43 histone tails and may associate with either transcription activation or repression. In
44 particular, the trimethylation of histone H3 at lysine 27 (H3K27me3),
45 monoubiquitination of histone H2A (H2Aub), and trimethylation of histone H3 at lysine 4
46 (H3K4me3) are hallmarks of chromatin regulation mediated by the *Polycomb* and
47 *Trithorax* group (PcG and TrxG) proteins, which are central to developmental regulation
48 [2,3].

49 In both animals and plants, H3K27me3 is the hallmark of the *Polycomb* Repressive
50 Complex 2 (PRC2) and represses the expression of target genes, including a range of key
51 transcriptional regulators across different developmental pathways [4–7]. PRC2-
52 mediated H3K27me3 plays essential roles in regulating cell and tissue identity, and
53 mutations of PRC2 components are characterized by phenotypes associated with loss
54 of cell identity and impaired cell proliferation [2,8–10].

55 PRC2 activity and H3K27me3 deposition are dynamically regulated in a tissue- and
56 developmental-stage dependent manner in both animals and plants [3,11,12]. H2Aub is
57 the hallmark of the *Polycomb* Repressive Complex 1 (PRC1) and plays a key role
58 collaborating with PRC2-H3K27me3, facilitating the repression of target genes. In
59 *Arabidopsis*, PRC1-mediated H2Aub functions upstream of PRC2, serving as a
60 prerequisite for H3K27me3 deposition [13–15]; accordingly, mutations in PRC1
61 components lead to loss of H3K27me3, derepression of PRC2 targets, and phenotypes
62 that resemble those of PRC2 mutants [14,15].

63 In contrast, genes key to development must be maintained active in the specific tissue
64 or developmental stage to support proper development. Antagonistic to PcG-mediated
65 repression, *Trithorax* group proteins deposit H3K4me3, which is crucial to maintain
66 active gene expression. Consistently, mutations in *Trithorax* group proteins lead to
67 developmental defects at various stages, including embryonic development and floral
68 transition [16–21].

69 Given their central roles in developmental regulation, comprehensive studies of PcG
70 and TrxG activities across the entire life cycle are key to understanding their functions.
71 Notably, the understanding of the dynamics of PcG and TrxG activities during
72 embryogenesis, an integral early stage of the plant life cycle, remains limited. The
73 embryogenesis of *Arabidopsis* is characterized by an early morphogenesis phase and a
74 later maturation phase [22]. One key question in embryogenesis is how the transition

from the morphogenesis phase to the maturation phase is regulated. Embryos enter the maturation phase at specific developmental stages, largely independent of the state of the endosperm and the uptake of sucrose, a major nutritional source [23]. These observations suggest that the onset of the maturation program is likely encoded in gene regulation programs in embryos rather than being determined by external factors. The LAFL transcription factors LEAFY COTYLEDON1 (LEC1), ABSCISIC ACID INSENSITIVE3 (ABI3), FUSCA3 (FUS3), and LEAFY COTYLEDON2 (LEC2) have been identified as master regulators of seed maturation, with gene regulatory networks between each LAFL factor and their downstream targets having been described to coordinate the maturation program [24,25]. However, how the morphogenesis-to-maturation transition is regulated at the chromatin level remains less explored, and the roles of PcG and TrxG proteins, which function as key regulators of other phase transitions [6,14,26,27], remain unclear in this context.

Loss-of-function mutation in *Polycomb* or *Trithorax* group proteins frequently leads to embryo lethality [28–32]. Similar embryo abortion phenotypes have also been observed in *Arabidopsis* for several *Polycomb* and *Trithorax* group proteins, notably for components of FIS-PRC2, a variant of PRC2 that functions mainly in early embryos [10,21,33]. Conversely, several other mutants in PcG proteins in *Arabidopsis* with severe defects in vegetative tissues exhibit only mild phenotypes during embryogenesis [8,34,35]. Therefore, *Arabidopsis* embryos appear relatively tolerant of mutations in *Polycomb* group proteins, making them a useful system to study PcG and TrxG functions during embryogenesis [10].

The understanding of PcG and TrxG activities in plant embryos is limited by the scarcity of epigenomic data in embryonic tissues [36]. By overcoming the material limitations of embryos using cleavage under targets and tagmentation followed by sequencing (CUT&Tag, [37,38]), we generated datasets of H3K4me3, H3K27me3, and H2Aub at two stages of *Arabidopsis* embryos that could provide insights into the chromatin dynamics during the transition between the morphogenesis and maturation phase. Our results revealed that H3K4me3 is associated with growth and cell cycle progression at the early stage of embryonic development, provided the first molecular evidence for H3K27me3 repression of the seed maturation program in the early embryos, and showed that H2Aub is associated with the active expression of seed development genes in embryos. Together, these findings provide valuable insights into the dynamic PcG and TrxG activities during embryogenesis and advance our understanding of their roles throughout the plant life cycle.

Results

Profiling of H3K4me3, H3K27me3, and H2Aub in *Arabidopsis* Embryos

We chose to profile the histone modifications at two developmental stages of embryos - the walking stick stage and the mature green stage, along with seedlings as a post-embryonic reference (**Figure 1a**). The walking stick stage is at the end of the morphogenesis phase, with the maturation programs already partially initiated, whereas the mature green stage represents a fully established maturation phase. Comparing these two stages may reveal shifts in chromatin dynamics during this developmental transition. Additionally, most epigenomic profiles of *Arabidopsis* have been derived from vegetative tissues such as seedlings [4,7]; therefore, comparing embryos with seedlings could uncover embryo-specific epigenetic features.

Embryo and seedling samples were used for CUT&Tag (Cleavage Under Targets and Tagmentation, [37,38]) to generate genome-wide profiles of H3K4me3, H3K27me3, and H2Aub (**Figure 1a, Supplementary Table S1**). Replicates show high similarity, datasets of different histones show a clear distinction (**Supplementary Figure S1**). For each histone modification, the datasets from two embryonic stages show the greatest similarity; the seedling datasets, despite clustering with the corresponding embryonic datasets, are more distant, reflecting developmental-stage-specific differences (**Figure 1b**).

These datasets provide a useful visualization of the genomic distribution of the histone modifications across developmental stages (**Figure 1c**). Histone-modification-enriched genomic regions (peaks) were identified with reliable quality metrics (**Supplementary Table S2**). For H3K27me3 and H3K4me3, approximately 10,000 peaks were identified in each developmental stage; whereas H2Aub yields fewer peaks, ranging from 800 to more than 3,000. A large number of peaks are shared across developmental stages, indicating that many histone-modification-associated regions are conserved. At the same time, subsets of stage-specific peaks were identified. Notably, we found that more than 1,000 peaks are unique to embryos for each histone modification, along with 200-2,000 seedling-specific peaks, suggesting features unique to embryos (**Figure 1d**).

In order to access the relationship of histone modifications and gene expression, we visualized the expression levels of genes associated with each histone modification using publicly available RNA-sequencing data (**Figure 1e**, [7,39,40]). We confirmed that H3K4me3 is associated with active gene expression, whereas H3K27me3 is associated with transcriptional repression. Although H2Aub is canonically considered a repressive mark, we observed an association of H2Aub with actively expressed genes, consistent with recent studies reporting H2Aub occupancy on actively transcribed genes [13,41–43].

Together, we generated high-quality datasets of H3K4me3, H3K27me3, and H2Aub in two developmental stages of embryos as well as seedlings of *Arabidopsis*, allowing a first comparative analysis of the dynamics of PcG and TrxG activities across embryos and vegetative tissues.

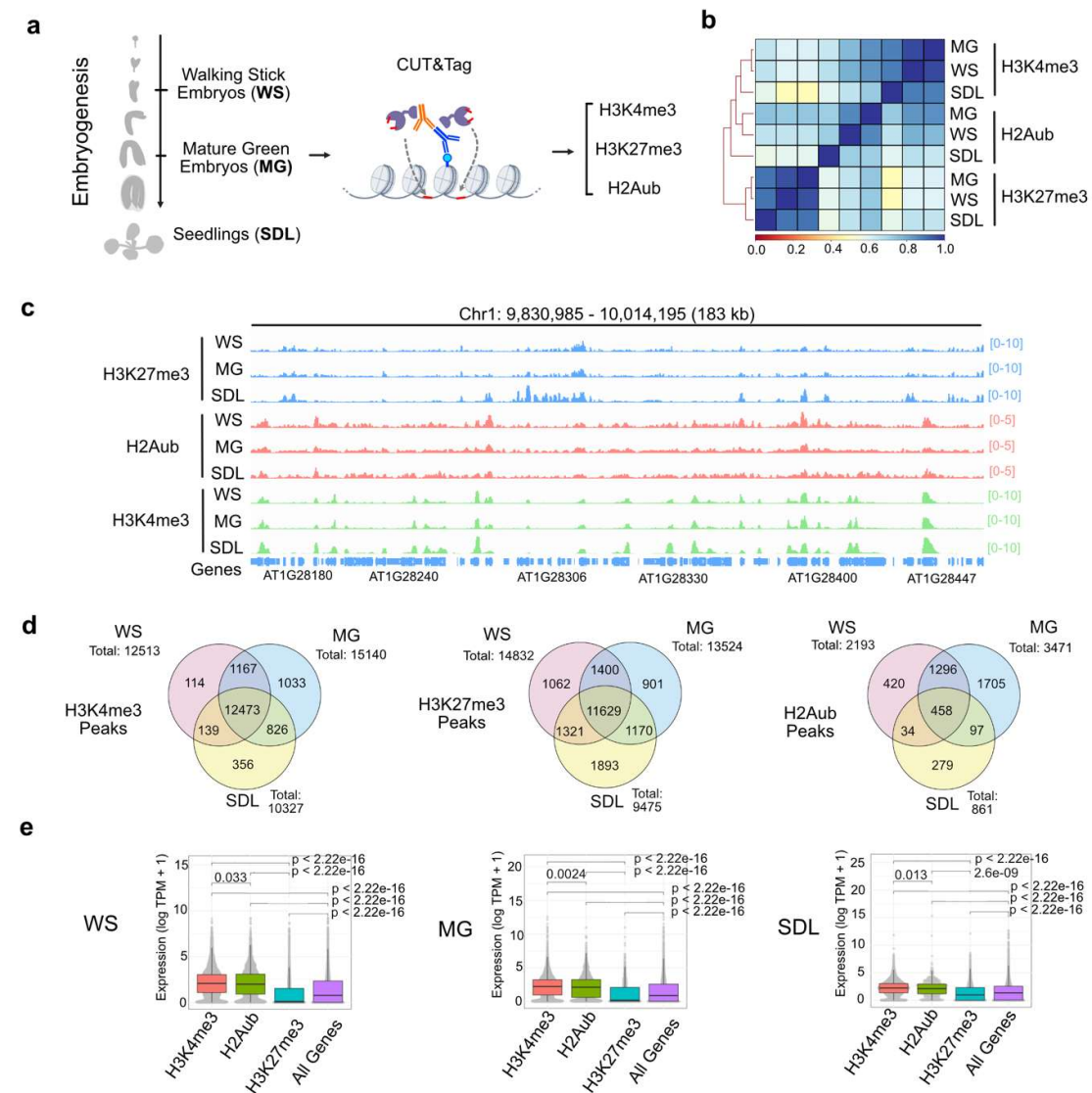


Figure 1. Profiling of H3K4me3, H3K27me3, and H2Aub in *Arabidopsis* Embryos.

(a) Graphic representation of the experiment procedures: samples from walking stick (WS) and mature green (MG) embryos and seedlings were collected and used for CUT&Tag to profile the genome-wide distribution of H3K4me3, H3K27me3, and H2Aub.

(b) Correlation matrix visualizing the similarities across developmental stages and histone modifications based on signal intensity from 500-bp genomic bins. The histone

160 modification signals have been normalized to the corresponding total H3 controls. WS:
161 walking stick embryos; MG: mature green embryos; SDL: seedlings.

162 (c) Integrative Genomics Viewer (IGV) screenshot of a 183-kb genomic window,
163 providing a visual overview of the datasets.

164 (d) Venn diagram visualizing the number of enriched genomic regions (peaks) identified
165 for each histone from each developmental stage, visualizing overlaps of histone-
166 modification-enriched regions across developmental stages.

167 (e) Box plot with jitter illustrating the expression levels of genes associated with each
168 histone modification at each developmental stage. Expression levels of all genes are
169 included for comparison. TPM: transcripts per million. Statistical significance was
170 assessed with wilcoxon rank sum test.

171

H3K4me3 is associated with growth and cell cycle progression at the earlier stage of embryonic development

We next examined the difference across developmental stages, starting with the active mark H3K4me3. Consistent across embryo and seedling samples, H3K4me3 shows enrichment around transcriptional starting sites (TSSs, **Figure 2a**). H3K4me3 in walking stick and mature green embryos are highly similar, yet remain distinguishable. Mature green embryos were used to compare with seedlings as they are closer in developmental timing, and the results show substantial differences in H3K4me3 between the two (**Figure 2b**). Motif enrichment analysis revealed motifs shared across developmental stages, as well as stage-specific motifs, such as HY5- and REF6- binding motifs in the walking stick embryos, MYB101-binding motif in mature green embryos, ASIL1- and RAV1- binding motifs in seedlings, suggesting potential developmental-stage-specific regulation mechanisms (**Supplementary Table S3**).

Differential analysis of H3K4me3 in walking stick and mature green embryos revealed around 400 genomic regions with stage-specific enrichment, with approximately 200 regions enriched in each stage (**Figure 2c**). Consistent with the epigenetic difference, genes associated with stage-specific H3K4me3 enrichment exhibit correspondingly higher expression in the same stage (**Figure 2d**).

To investigate the functions of genes with stage-specific enrichment, we conducted gene ontology (GO) enrichment analysis. Genes with elevated H3K4me3 levels in the walking stick embryos were strongly enriched for GO terms related to growth and cell cycle processes (**Figure 2e**). Representative examples include: *PIN-FORMED 6* (*PIN6*, *AT1G77110*, **Figure 2f**), a gene annotated with the GO term “growth”; *CYCLIN D3;2* (*CYCD3;2*, *AT5G67260*) and *CYCLIN B1;4* (*CYCB1;4*, *AT2G26760*) (**Supplementary Figure S2**), two genes annotated with the GO term “cell cycle progress”. These results highlight the roles of H3K4me3 in regulating cell division and growth during embryogenesis.

To reveal epigenetic signatures specific to embryos, we compared embryos with seedlings (**Supplementary Figure S3**). Genes associated with elevated H3K4me3 in walking stick embryos were enriched for GO terms related to plant organ formation (**Figure 2g**), whereas those in mature green embryos were primarily enriched for GO terms related to the seed maturation process (**Figure 2h**). The results highlight that the stage-specific developmental programs are accompanied by coordinated changes in the H3K4me3 distribution.

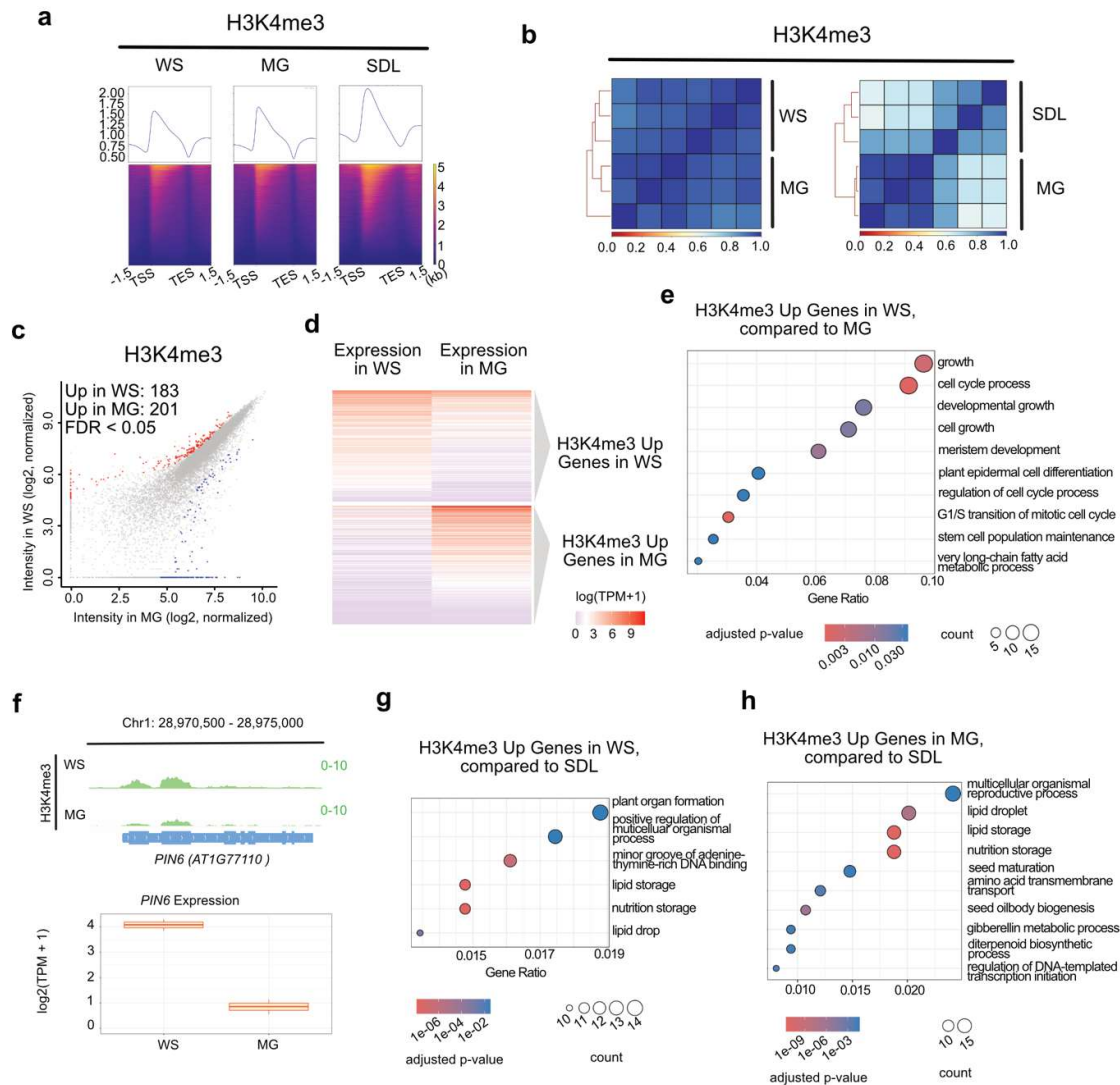


Figure 2. H3K4me3 is associated with growth and cell cycle at the early stage of embryonic development.

(a) Genomic heatmaps visualizing the distribution of H3K4me3 along gene bodies. WS: walking stick embryos; MG: mature green embryos; SDL: seedlings.

(b) Correlation matrices visualizing H3K4me3 similarity between walking stick and mature green embryos, and between mature green embryos and seedlings.

Comparisons were based on signal intensity within H3K4me3 peak areas calculated with DESeq2. All three biological replicates per stage were included.

(c) Scatter plot comparing H3K4me3 levels in the walking stick embryos and mature green embryos. Each dot represents an H3K4me3-enriched genomic region. Blue and red dots indicate regions with significant differences between the two developmental stages.

220 (d) Heatmap visualizing the expression levels of genes associated with differential
221 H3K4me3 enrichment between walking stick embryos and mature green embryos.

222 (e) Gene ontology (GO) enrichment analysis visualizing the functional enrichment of
223 genes with significantly elevated H3K4me3 levels in walking stick embryos relative to
224 mature green embryos.

225 (f) Upper: Integrative Genomics Viewer (IGV) screenshots showing H3K4me3 levels of a
226 growth-related gene, *PIN6* (*AT1G77110*), in walking stick embryos compared with
227 mature green embryos. Lower: corresponding expression levels of *PIN1* in walking stick
228 embryos compared with mature green embryos. TPM: transcripts per million.

229 (g) GO enrichment analysis visualizing the functional enrichment of genes with
230 significantly elevated H3K4me3 levels in walking stick embryos compared with
231 seedlings.

232 (h) GO enrichment analysis visualizing the functional enrichment of genes with
233 significantly elevated H3K4me3 levels in mature green embryos compared with
234 seedlings.

235

H3K27me3 represses the seed maturation program in embryos at the early stage

We next examined the distribution of H3K27me3, the repressive mark deposited by the *Polycomb* Repressive Complex 2 (PRC2). In all developmental stages, H3K27me3 is predominantly enriched on gene bodies (**Figure 3a**). H3K27me3 distribution shows a high degree of similarity between walking stick and mature green embryos, but is clearly distinct in seedlings (**Figure 3b**).

Differential analysis of H3K27me3 between walking stick and mature green embryos revealed approximately 200 genomic areas with elevated H3K27me3 in the walking stick stage, whereas only one region showed significantly higher H3K27me3 in the mature green stage (**Figure 3c**). This asymmetry suggests that during cell proliferation in embryos, H3K27me3 is preferentially lost rather than newly established. Notably, though, the single region with significantly increased H3K27me3 in walking stick embryos is associated with two genes, *COLD SHOCK DOMAIN PROTEIN 4* (*ATCSP4*, *AT2G21060*) and *LIKE AUXIN RESISTANT 2* (*LAX2*, *AT2G21050*), both of which are annotated with GO terms “embryo development” (**Supplementary Table S4**) and show higher expression at the earlier stage (**Figure 3d**). This observation suggests that, despite being rare, H3K27me3 deposition during embryogenesis likely contributes to developmental regulation.

Consistent with the repressive role of H3K27me3, the genes with higher H3K27me3 at the walking stick stage show reduced expression at that stage (**Figure 3d**). Gene ontology (GO) enrichment analysis found significant enrichment for several GO terms related to the seed maturation process (**Figure 3e**, **Supplementary Table S4**). These results support a model that H3K27me3 represses the seed maturation program in embryos at the early stages and is later removed, thereby allowing its activation at the appropriate time.

The seed development genes represent a major class of PRC2 targets and H3K27me3-associated loci in vegetative tissues [7], and the current model suggests that the H3K27me3 deposition on these genes occurs during the seed-to-seedling transition [6]. Consistent with this model, seed maturation genes are depleted in H3K27me3 in mature green embryos relative to seedlings (**Figure 3f**, **Supplementary Figure S4**, **Supplementary Table S4**). Together, the results revealed dynamic H3K27me3-mediated regulation of seed maturation genes in embryos: repression during early embryogenesis, derepression at late embryogenesis to permit a temporal window of expression and function, and re-establishment upon transition to seedlings. *SEED STORAGE ALBUMIN 3* (*AT2S3*, *AT4G27160*, **Figure 3g**) and *DELAY OF GERMINATION 1* (*DOG1*, *AT5G45830*, **Supplementary Figure S5**) represent two examples of this gene class.

The *LAFL* genes, comprising *LEAFY COTYLEDON 1* (*LEC1*, AT1G21970), *ABA*
INSENSITIVE 3 (*ABI3*, AT3G24650), *FUSCA3* (*FUS3*, AT3G26790), *LEAFY COTYLEDON 2*
(*LEC2*, AT1G28300), encode four master transcriptional regulators that regulate
differentiation and the onset of seed maturation [22,24,44]. Notably, they show among
the most significant differences in H3K27me3 between embryos and seedlings (**Figure**
3h, Supplementary Figure S5, Supplementary Table S5), highlighting that genes with
key developmental functions are under particularly stringent H3K27me3-mediated
regulation. Furthermore, although expressions of *LEC1* and *LEC2* have largely ceased at
the mature green stage (**Figure 3h, Supplementary Figure S5**), minimal changes of
H3K27me3 were observed on these loci, suggesting that substantial establishment of
H3K27me3 occurs only upon transition to seedlings.

In addition to regulating seed maturation programs in embryos, our analysis suggests
that H3K27me3 may be involved in other biological processes. Comparison of
H3K27me3 between walking stick embryos and seedlings revealed a downregulation of
H3K27me3 at genes related to saccharide transportation pathways in embryos
(**Supplementary Figure S4**). These include *SWEET11* and *SWEET12*, two sucrose
transporters highly expressed in seeds. This observation suggests a potential role of
H3K27me3 in regulating nutritional uptake in embryos [23].

Finally, around 2,800 to 3,700 genomic areas in walking stick and mature green embryos
are occupied by both H3K4me3 and H3K27me3 (**Supplementary Figure S6**). Genes
associated with both marks in walking stick embryos show an increased expression at
the mature green stage and are enriched for seed maturation functions, whereas genes
associated with both marks in the mature green stage are enriched for functions related
to germination (**Supplementary Figure S6**). These findings suggest genes associated
with both H3K4me3 and H3K27me3 are enriched for functions required at subsequent
developmental stages, which is similar to observations in plant shoot meristems and
sperm cells, and may represent a common feature in plant development [40,45].

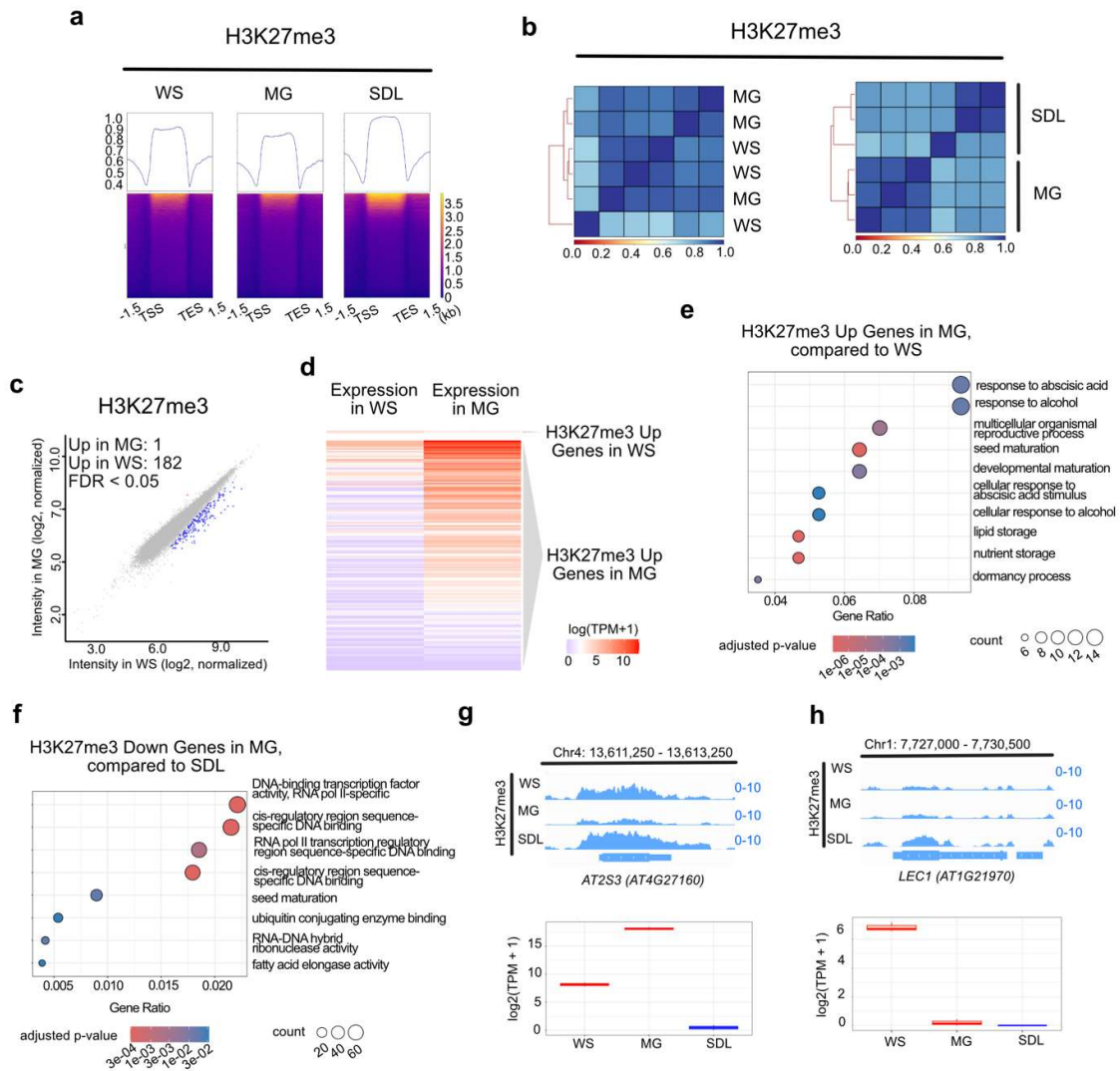


Figure 3. H3K27me3 represses the seed maturation program in embryos at the earlier stage.

(a) Genomic heatmaps visualizing the distribution of H3K27me3 along gene bodies. WS: walking stick embryos; MG: mature green embryos; SDL: seedlings.

(b) Correlation matrices visualizing H3K27me3 similarity between walking stick and mature green embryos, and between mature green embryos and seedlings. Comparisons were based on signal intensity within H3K27me3 peak areas calculated with DESeq2. All three biological replicates per stage were included.

(c) Scatter plot comparing H3K27me3 levels in the walking stick embryos and mature green embryos. Each dot represents an H3K27me3-enriched genomic region. Blue and red dots indicate regions with significant differences between the two developmental stages.

316 (d) Heatmap visualizing the expression levels of genes associated with differential
317 H3K27me3 enrichment between walking stick embryos and mature green embryos.

318 (e) Gene ontology (GO) enrichment analysis visualizing the functional enrichment of
319 genes with significantly elevated H3K27me3 levels in walking stick embryos relative to
320 mature green embryos.

321 (f) GO enrichment analysis visualizing the functional enrichment of genes with
322 significantly lower H3K27me3 levels in mature green embryos compared with seedlings.

323 (g) Upper: Integrative Genomics Viewer (IGV) screenshots showing H3K27me3 levels of
324 a seed-maturation-related gene, *STORAGE ALBUMIN 3* (*AT2S3*, *AT4G27160*), in walking
325 stick and mature green embryos compared with seedlings. Lower: corresponding
326 expression levels of *AT2S3* in walking stick and mature green embryos compared with
327 seedlings. TPM: transcripts per million.

328 (h) Upper: Integrative Genomics Viewer (IGV) screenshots showing H3K27me3 levels of
329 a *LAFL* transcription factor, *LEAFY COTYLEDON 1* (*LEC1*, *AT1G21970*), in walking stick
330 and mature green embryos compared with seedlings. Lower: corresponding expression
331 levels of *LEC1* in walking stick and mature green embryos compared with seedlings.
332 TPM: transcripts per million.

333

334

H2Aub is associated with active expression of seed development genes in embryos

Finally, we examined the dynamics of the H2Aub mark. The H2Aub mark is enriched on gene bodies, with a slight bias toward transcriptional start sites (TSSs) (**Figure 4a**). Similar to those of H3K4me3 and H3K27me3, H2Aub profiles in two embryonic stages show high similarity, whereas differences between mature green embryos and seedlings were observed (**Figure 4b**). Nevertheless, comparisons between two embryonic stages have identified nearly 2,000 peaks specific to the mature green stage and nearly 500 specific to the walking stick stage, suggesting stage-specific features (**Supplementary Figure S7**).

Differential analysis identified thirty genomic regions with significantly enriched H2Aub in the mature green stage, and two in the walking stick stage (**Figure 4c**). Genes associated with enriched H2Aub in the mature green stage also exhibit higher expression in that stage (**Figure 4d**). While Figure 1e indicates an association between H2Aub presence and higher gene expression, the findings here further suggest that increased H2Aub levels are accompanied by higher gene expression. These results strengthen the link between H2Aub and active gene expression in *Arabidopsis* [13,41–43]. Despite this association, H2Aub is the hallmark of *Polycomb* Repressive Complex 1 (PRC1) and is suggested to recruit PRC2 to deposit H3K27me3 in *Arabidopsis* [13]. Consistent with this model, H3K27me3 peaks have been found on nearly half of H2Aub peaks across developmental stages, and genes associated with both marks exhibit lower expression levels relative to those marked by H2Aub alone (**Supplementary Figure S7**).

To explore the functions of these genes associated with H2Aub in embryos, we tested their gene ontology enrichment (**Figure 4e**, **Supplementary Table S6**). Notably, enriched GO terms included processes related to seed development, such as cotyledon development and seed maturation. Widespread H2Aub peaks have been found on several well-known seed-development genes, such as *LEAFY COTYLEDON 1* (*LEC1*, *AT1G21970*) and *SEED STORAGE ALBUMIN 3* (*AT2S3*, *AT4G27160*) (**Figure 4f**). Although these genes have different expression patterns in embryos, *LEC1* is highly expressed at the walking stick stage, and *AT2S3* is highly expressed at the mature green stage; the presence of H2Aub on these genes has been detected at similar levels in both stages. These observations support a model in which H2Aub's association with actively expressed genes contributes to the regulation of seed development.

While we observed H2Aub occupancy on seed development genes in embryos, previous studies have shown that H2Aub recruits PRC2 to deposit H3K27me3 and establish repression on these genes during the seed-to-seedlings transition [14]. To further explore the link between H2Aub occupancy and subsequent H3K27me3 acquisition, we intersected H2Aub-associated genes in mature green embryos with

genes that showed increased H3K27me3 levels in seedlings compared to embryos (Figure 4g). The intersection revealed 526 genes in overlap, with a significantly enriched set of 40 genes related to seed development (Figure 4h). Thus, the results support a model in which H2Aub marks actively expressed seed development genes in embryos and subsequently serves as a platform for PRC2 recruitment and H3K27me3 deposition during the seed-to-seedling transition. This suggests a revision to the previous model, which proposed that H2Aub initiates H3K27me3 establishment upon the transition to seedlings [14].

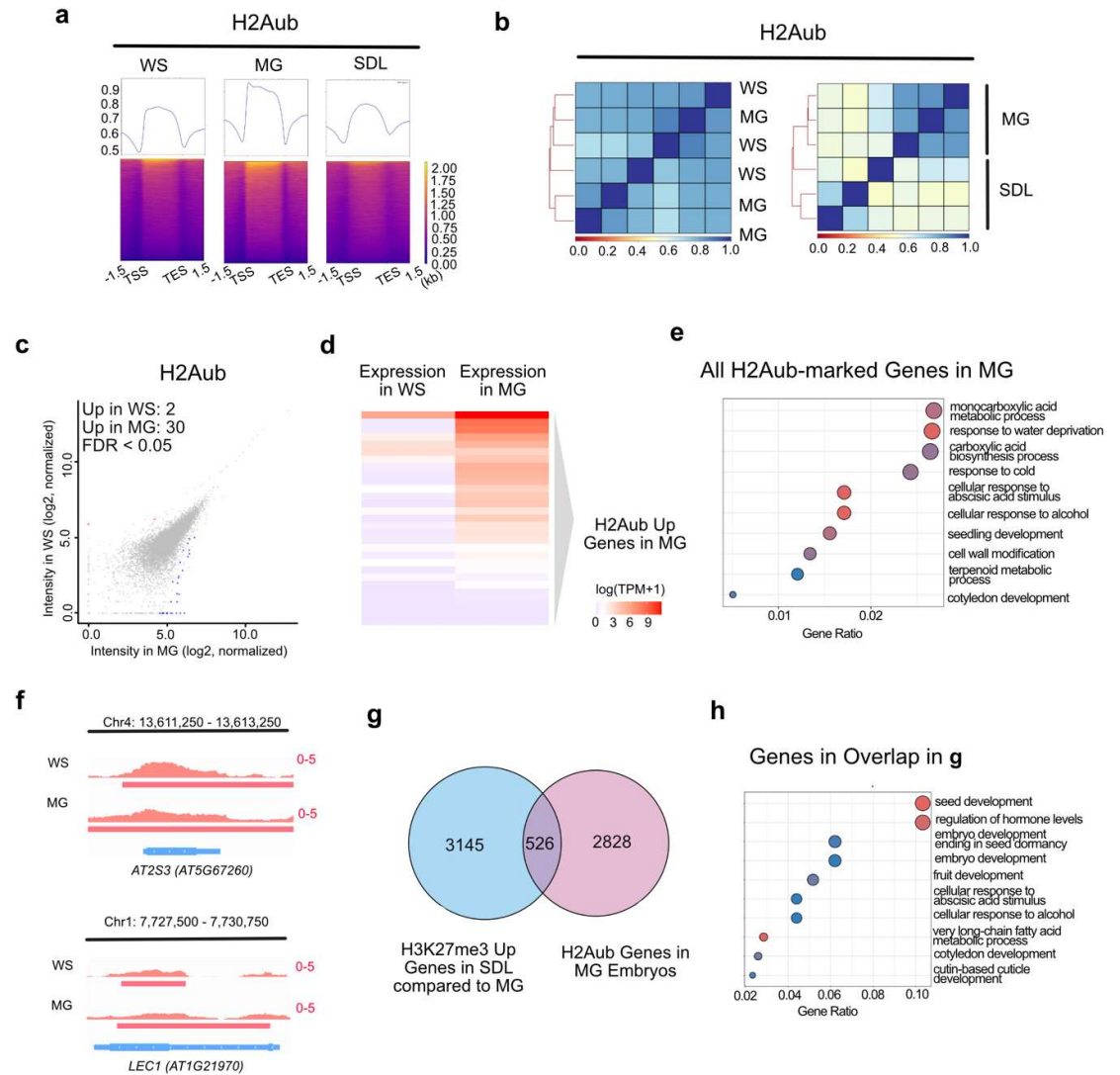


Figure 4. H2Aub is associated with active expression of seed development genes in embryos.

(a) Genomic heatmaps visualizing the distribution of H2Aub along gene bodies. WS: walking stick embryos; MG: mature green embryos; SDL: seedlings.

(b) Correlation matrices visualizing H2Aub similarity between walking stick and mature green embryos, and between mature green embryos and seedlings. Comparisons were based on signal intensity within H2Aub peak areas calculated with DESeq2. All three biological replicates per stage were included.

(c) Scatter plot comparing H2Aub levels in the walking stick embryos and mature green embryos. Each dot represents an H2Aub-enriched genomic region. Blue and red dots indicate regions with significant differences between the two developmental stages.

(d) Heatmap visualizing the expression levels of genes associated with differential H2Aub enrichment between walking stick embryos and mature green embryos.

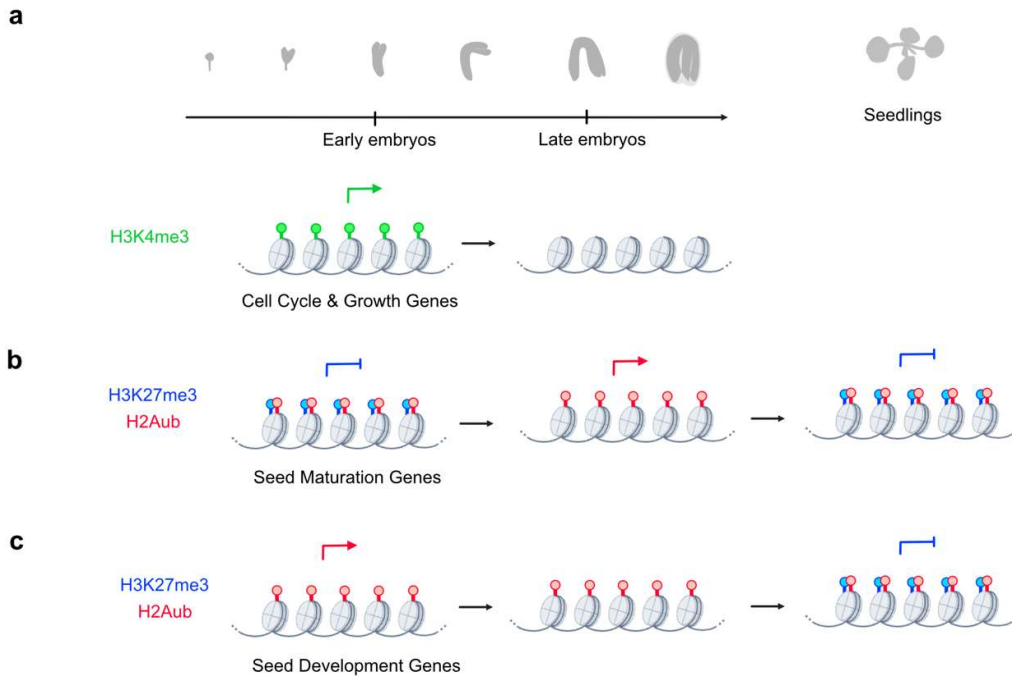
(e) Gene ontology (GO) enrichment analysis visualizing the functional enrichment of all H2Aub-marked genes in mature green embryos.

(f) Integrative Genomics Viewer (IGV) screenshots showing H2Aub levels of two seed-development-related genes, *LEC1* (*AT1G21970*) and *AT2S3* (*AT4G27160*), in embryos, with enriched regions (peaks) annotated. Expression levels for these genes are shown in Figures 3g and 3h.

(g) Venn diagram displaying the overlap of H2Aub-associated genes in mature green embryos, with genes that have elevated levels of H3K27me3 in seedlings compared to mature green embryos.

(h) GO enrichment analysis visualizing the functional enrichment of the genes in the overlap shown in (g).

410 **Discussion**



411
412 **Figure 5.** Working models of H3K4me3, H3K27me3, and H2Aub in *Arabidopsis* embryos

413 (a) Working model of H3K4me3 dynamics at cell cycle- and growth-related genes across
414 embryos and seedlings. H3K4me3 enrichment at these genes in early embryos allows
415 their expression and promotes growth and cell proliferation.

416 (b) Working model of H3K27me3 and H2Aub dynamics at seed maturation genes across
417 embryos and seedlings. Seed maturation genes are initially marked by both H2Aub and
418 H3K27me3 in early embryos; removal of H3K27me3 in later embryos creates an H2Aub-
419 only state, allowing expression. During the transition to seedlings, H3K27me3 is
420 deposited at H2Aub-marked loci to re-establish repression.

421 (c) Working model of H3K27me3 and H2Aub dynamics at seed development genes
422 other than the seed maturation genes across embryos and seedlings. H2Aub
423 occupancy at seed development genes is associated with their active expression in
424 embryos. During the transition to seedlings, H3K27me3 is deposited at H2Aub-marked
425 loci to re-establish repression.

427

428 To elucidate the dynamics of *Polycomb* and *Trithorax* group proteins in *Arabidopsis*
429 embryos, we generated genome-wide profiles of H3K4me3, H3K27me3, and H2Aub.
430 Our results addressed a key gap in epigenetic knowledge across the plant life cycle and
431 uncovered key pathways regulated by *Polycomb* and *Trithorax* group proteins during
432 embryogenesis. Furthermore, our results revealed a non-canonical association of
433 H2Aub with a broad set of seed development genes in embryos, thereby revising the
434 previous model of H2Aub function in mediating the seed-to-seedling transition [14].

435 Our results revealed an important role of H3K4me3 in regulating growth and cell cycle
436 progression in the early embryos (**Figure 5a**). Loss-of-function mutations of *Trithorax*
437 group genes frequently result in growth defects [20], including developmental arrest at
438 early stages of embryogenesis [21]. Studies have shown that H3K4me3 is not only
439 correlated with active gene expression but also instructive to active transcription
440 [46,47]. In combination with these reports, our results support the model that H3K4me3
441 deposition promotes cell proliferation and growth in early embryos. While studies of
442 *Trithorax* group proteins have largely emphasized their roles in cell identity regulation,
443 our results highlight that, like the *Polycomb* group proteins [10], they also play
444 important roles in cell proliferation regulation.

445 We found that H3K27me3 is enriched at seed maturation genes in early embryos and
446 depleted from these genes at later stages (**Figure 5b**). While seed maturation is a gene
447 ontology term covering processes in the whole seed, including embryos, endosperm,
448 and seed coat, we specifically examined the dynamics of genes with this GO term in
449 embryos. In vegetative tissues, the seed maturation genes constitute a major category
450 of PRC2 target genes and have served as models for investigating gene repression
451 mediated by PcG proteins [6,7,48]. Repression states on these genes are established
452 during the seed-to-seedling transition [6,14]. Although most seed maturation genes only
453 begin to express after a certain stage of embryogenesis [23], previous reports have
454 assumed that they are activated by transcriptional activators [25], whereas the
455 possibility of repressive regulation has been overlooked. Our findings established PRC2-
456 mediated repression as a key regulator of maturation programs in embryos and
457 advanced our understanding of the regulation of the onset of the maturation process.

458 The discovery of H3K27me3 repression of seed maturation programs in early embryos
459 raises several open questions. One important question is how H3K27me3 is removed
460 from these genes. H3K27me3 can be removed either through active demethylation or
461 passive dilution during cell division [49]. The highly dynamic, rapidly developing
462 embryos constitute a unique system for studying the maintenance and removal of
463 histone marks, with the potential to provide novel insights into epigenetic mechanisms
464 mediated by histone modifications.

Another question concerns how the specificity of H3K27me3 removal is achieved during embryogenesis. We have observed that the majority of H3K27me3 peaks remain stable from earlier to later embryos, while a group of around 200 peaks shows a significant reduction. The specificity of PRC2 recruitment in plants is mediated by a range of recruiter motifs and transcription factors [48,50,51]. Accordingly, the underlying motif composition in these peaks may also contribute to determining H3K27me3 removal. A candidate motif is the RY-rich motif, recognized by the VAL family proteins, known PRC2 recruiters [51], as well as by the B3 family proteins, including key regulators of embryogenesis [52]. Investigating this question could provide valuable insights into the mechanism of H3K27me3 removal.

Finally, we showed that H2Aub is associated with actively expressed seed development genes in embryos (**Figure 5b, 5c**). As in the case of seed maturation genes, seed development is a GO term covering processes in whole seeds, and we specifically examined the dynamics of genes with that GO term in embryos. While studies have shown an association between H2Aub and gene activation [13,41–43], whether this non-canonical role contributes to the developmental programs remains unknown. By showing that H2Aub levels positively correlate with expression across stages of embryogenesis and are associated with seed developmental programs in embryos, we provide evidence for H2Aub's role in activating seed development genes in embryos. This observation raises the question of whether H2Aub plays a causal role in the active expression of seed development genes in embryos, which could be addressed by analyzing their expression in embryos of PRC1 mutants.

By revealing H2Aub's presence on seed development genes in embryos, we extended the previous model of H2Aub function in mediating seed-to-seedling transition [14]. Our results indicate that H2Aub acts as a prerequisite, rather than a direct signal, for subsequent deposition of H3K27me3 on seed development genes in embryos during the transition to seedlings. As a result, the presence of H2Aub at seed development genes in both embryos and seedlings indicates that additional, yet unidentified factors contribute to the distinct H3K27me3 states in embryos and seedlings. Investigating this question may therefore provide valuable insights into PRC2 recruitment mechanisms.

These datasets provide additional opportunities to explore potential regulatory mechanisms during embryogenesis, such as the potential regulation of nutritional uptakes in embryos by PRC2 (Supplementary Figure S4), and a potential association of genomic regions where H3K27me3 and H3K4me3 are both detected, and genes functioning in the subsequent developmental stages (Supplementary Figure S6). As a highly dynamic developmental process, embryogenesis provides an ideal platform to address these questions and holds the potential to yield novel biological insights into gene regulation by the *Polycomb* and *Trithorax* group proteins, broadly applicable to other developmental processes.

From a technical standpoint, this study demonstrated the strength of CUT&Tag in revealing epigenetic states in small and rare tissues. Epigenetic profiling from limited material has been challenging, as conventional chromatin immunoprecipitation sequencing typically requires a gram-level sample [37,38], limiting the availability of epigenomic datasets for plant embryos. Our study demonstrated that tissue-specific epigenomic profiles are valuable for dissecting epigenetic regulation and are expected to encourage future studies of tissue-specific epigenetic regulation.

In summary, we revealed the states of H3K4me3, H3K27me3, and H2Aub in two developmental stages of *Arabidopsis* embryos. The results identified multiple pathways regulated by these epigenetic marks, provided new insights into gene regulation by the *Polycomb* and *Trithorax* group proteins, and prompted many important open questions. Furthermore, the datasets provide valuable resources for researchers interested in the chromatin regulation of diverse pathways in embryos, may inspire future studies of tissue-specific epigenetic states, and could support future engineering efforts in seed development.

Conclusions

Epigenomic profiles in *Arabidopsis* embryos revealed that H3K4me3 is associated with growth and cell cycle progression at the earlier stage of embryonic development, H3K27me3 represses the seed maturation program in embryos at the early stage, and H2Aub is associated with active expression of seed development genes in embryos. The results highlight the roles of *Polycomb* and *Trithorax* group proteins in regulating growth, cell proliferation, development, and developmental phase transitions.

Competing interests

The authors declare that there are no competing interests.

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Authors' contributions

SS conceived the idea, provided supervision, analyzed the data, and acquired funding; YF performed the experiments and analyzed the data; MS contributed to the experiments; YF and SS wrote the manuscript.

538

539 **Availability of data and materials**

540 Raw sequencing data have been deposited in the European Nucleotide Archive under
541 accession number PRJEB108232. The scripts used for analysis are available at the
542 GitHub repository: [https://github.com/yixuan-](https://github.com/yixuan-fu22/Polycomb_Trithorax_Dynamics_in_Arabidopsis_Embryos)
543 [fu22/Polycomb_Trithorax_Dynamics_in_Arabidopsis_Embryos](https://github.com/yixuan-fu22/Polycomb_Trithorax_Dynamics_in_Arabidopsis_Embryos)

544

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559

560 **Methods**

561 **Plant material and growth condition**

562 *Arabidopsis thaliana* wild-type Columbia-0 (Col-0, NASC #N70000) seeds were used for
563 this study. Seeds were sown on half-strength MS media (1/2 MS salt base [Carolina
564 Biologicals, USA], 1% sucrose, 0.05% MES, 0.8% Phytoagar [Duchefa], pH>5.7 adjusted
565 with KOH), stratified in darkness at 4°C for 1-3 days, then grown under long-day
566 conditions (8h dark at 18°C, 16h light at 22°C, 70% relative humidity) in a plant growth
567 chamber (CU36L6, CLF Plant Climatics). Seedlings used for CUT&Tag material were
568 harvested between one and two weeks after starting incubation, when they had
569 developed 4-8 true leaves. Plants used for embryo isolation are transferred to soil and
570 grown under long-day conditions until flowering and seed set.

571

572 **Embryo isolation**

Developing siliques collected from flowering *Arabidopsis* plants were dissected under a stereomicroscope using a pair of syringes. Because seeds within the same silique are typically synchronized in developmental stage [53], two to three seeds from each silique were first opened to determine developmental stage based on morphology. Once the desired stage was identified, the remaining developing seeds were collected into a droplet of ice-cold PBS 1X (supplemented with the Protease Inhibitor Cocktail (cOmplete, EDTA-free Protease Inhibitor Cocktail, Roche #6417810)) in a shallow rectangular container (lid of an 8-well Tissue Culture Chambers REF94.6190.802, SARSTEDT). The seeds were then covered with a coverslip and pressed gently with a tweezer to release the embryos. The coverslip was then removed using a tweezer and washed with 1.5 mL of fresh ice-cold 1X PBS, which was collected into the samples to form a suspension containing embryos and tissue debris. The embryos were carefully isolated using 20 μ L and 200 μ L pipettes and transferred to a clean drop of PBS 1X. The developmental stage of each embryo was reassessed during collection to ensure that only embryos at the desired stage were included.

Liquid in the droplet was carefully removed using 200 μ L and 20 μ L pipettes, then 200 μ L of fresh 1X PBS was added for washing. Each sample was washed at least twice until no debris was visible under the microscope. The isolated embryos were maintained in a droplet of PBS 1X and stored at -80°C until use. Approximately 360 walking stick embryos or 200 mature green embryos were collected for each sample in the CUT&Tag experiment.

Cleavage Under Targets and Tagmentation (CUT&Tag)

CUT&Tag followed the previously published protocol [38]. Antibodies used were Rabbit anti-H3K27me3 (Cell Signaling Technology #9733, 1:50 dilution), Rabbit anti-H3K4me3 (Diagenode c15410003, 1:75 dilution), Rabbit anti-Ubiquityl-Histone H2A (Cell Signaling Technology #8240, 1:25 dilution), Mouse anti-H3 (Active Motif 39064, 1:75 dilution), Guinea Pig anti-Rabbit IgG (antibodies-online #ABIN101961, 1:50 dilution), and Rabbit anti-Mouse IgG (Abcam 46540). For each histone modification sample, the nuclei were split evenly into parallel tubes for the histone modification assay and total H3 control, to ensure that histone modification and control libraries are generated from the same batch of nuclei. Sequencing libraries were pooled and sequenced on Illumina NovaSeq X Plus Platforms that produced 150bp pair-ended sequences, with a targeted sequencing depth of 20-50 million read pairs per library. All CUT&Tag experiments were performed in three biological replicates, except for seedling H2Aub datasets, where seven replicates were generated and merged into three groups to account for lower unique read count.

Analysis of CUT&Tag data

Sequencing reads were quality-checked using fastQC (Version 0.12.1 [54]) and trimmed with fastp (Version 0.23.4 [55]) with options “fastp -w 4 -l 30 --trim_poly_x --poly_x_min_len 32”. Trimmed reads were aligned to the *Arabidopsis* TAIR10 reference genome using Bowtie2 (Version 2.4.1 [56]) with parameters “--local --very-sensitive --no-mixed --no-discordant --phred33 -l 10 -X 700”. Reads with mapping quality (MAPQ) scoring less than 10 were filtered out using Samtools (Version 1.17 [57]) with the option “samtools view -q 10”. Duplicate reads were marked using Picard tools (version 3.1.1, broadinstitute.github.io/picard). Sequencing depth, number of aligned reads, and number of unique reads were obtained using Samtools (samtools view -c) to evaluate alignment quality and duplication rates.

Peak Calling and Annotation

Peak calling was performed with MACS2 (Version 2.2.9.1 [58]) with options -f BAMPE --nomodel -q 0.05 --broad --broad-cutoff 0.1 -B --SPMR, total H3 samples as controls. Only nonduplicated reads were used for peak calling. To calculate the Fraction of Reads in Peaks (FRiP) score, the .bam files after duplication removal were transformed to .bed files (bedtools bamtobed), and the number of reads in peaks was obtained by intersecting the .bed files with peak files using bedtools (Version 2.31.0, [59]) [60].

Peaks detected in at least two of the three replicates were identified as consensus peaks. Consensus peaks were extracted using BEDTools (bedtools intersect) and used for subsequent analysis, except for seedling H2Aub datasets, where the peaks from the replicates with the highest nonduplicated reads were used to account for lower read counts.

Peak annotation was conducted using the flankgeneID feature of ChIPseeker (Version 1.46.1, [61]). Parameters used are: tssRegion=c(-500, 0), addFlankGeneInfo = TRUE, flankDistance = 500), such that peaks were annotated with genes located within 500 bp distances.

Visualizations

To visualize the genomic distribution of histone modifications, .bedGraph files were generated by MACS2 during peak calling with the “-B --SPMR” flag. The .bedGraph files were then background-corrected by their corresponding total H3 controls using MACS2 (MACS2 bdgcmp), converted to .bw files using bedGraphToBigWig (version 445, [62]) with the FE (Fold Enrichment) option, and visualized with the Integrative Genomics Viewer (IGV, version 2.19.7 [63]).

To visualize the distribution of histone marks along gene bodies, genomic heatmaps were generated using deepTools computeMatrix (scale-regions) and plotHeatmap (Version 3.5.3, [64]).

To assess sample similarity, background-corrected bigwig files were analyzed with deepTools (multiBigwigSummary) using either 500-bp genomic bins (--binSize 500) or peak areas (--BED). The options used are provided in the figure legends. The differences between samples and controls were analyzed using the sequencing-depth-normalized bigwig file generated by deepTools (bamcoverage) (Figure S1).

Overlap of gene sets was visualized using the R package VennDiagram (Version 1.8.2, [65], R version: 4.5.2). Overlap of peaks was first calculated using the R package GenomicRanges (Version 1.62.1, [66], R version: 4.5.2) and visualized using VennDiagram.

Analysis of RNA-seq data

Public RNA sequencing data from *Arabidopsis* seedlings and embryos were downloaded from the European Nucleotide Archive (accession numbers: ERR10163199, ERR10163200 [40]) and NCBI Sequence Read Archive (accession numbers: SRR8054371, SRR8054372, SRR8054373, SRR8054377, SRR8054378, SRR8054379 [39]; SRR2500947, SRR1931614 [7]). Reads were quality-checked using fastqc (0.12.1, [54]) and trimmed with fastp (version 0.23.4, -w 16 -l 30 --trim_poly_x --poly_x_min_len 32, [55]).

Kallisto (version 0.48.0, [67]) was used to pseudo-align the trimmed reads to the reference transcriptome. Transcripts per million (TPM) numbers obtained from Kallisto output were read and normalized using Sleuth (version 0.30.1, [68]) to obtain normalized transcriptome data.

Gene expression levels were visualized using box plots and heat plots made with the R package ggplot2 (version 4.0.1, [69], R version: 4.5.2). Jittered data points were added to boxplots using ggbeeswarm (version 0.7.3, [70], R version: 4.5.2). For comparing expression levels of genes marked by different histone modifications, statistical comparisons were performed using Wilcoxon rank sum test from the R package ggpubr (version 0.6.2, stat_compare_means(method = "wilcox.test", paired = FALSE), [71], R version: 4.5.2).

Differential Enrichment Analysis

Differential enrichment analysis of histone modifications was performed with the R package DiffBind (Version 3.20.0, [72], R version: 4.5.2, method=DBA_DESEQ2), which

implements DESeq2 (Version 1.50.2, [73]) for normalization and differential testing. Identified differential enriched regions were subsequently annotated using the flankgeneID feature of ChIPseeker (Version 1.46.1, [61], tssRegion=c(-500, 0), addFlankGeneInfo = TRUE, flankDistance = 500). Comparisons were visualized using XY scatter plots generated with ggplot2.

Gene Ontology Enrichment Analysis

Gene Ontology (GO) Enrichment Analysis was performed using the R package clusterProfiler (Version: 4.18.4, [74], R Version: 4.5.2, pAdjustMethod = "BH", qvalueCutoff = 0.05) and annotation database org.At.tair.db (Version 3.22.0, [75]). For gene sets identified by differential analysis, the background universe for GO enrichment tests was defined as all genes associated with the histone modification of interest in at least one developmental stage.

Motif Enrichment Analysis

Motif enrichment analysis was performed by MEME Suite (Version 5.4.1, [76]). BED files containing consensus peaks were converted to FASTA sequences using the bed2fasta function. Motifs enrichment was tested using meme-chip (-order 2 -minw 6 -maxw 15 -meme-nmotifs 10 -meme-searchsize 100000 -streme-pvt 0.05 -streme-align) and JASPAR 2022 Core Plant non-redundant database [77]. Motifs that were repetitive or did not match any known transcription factors were excluded from the results.

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